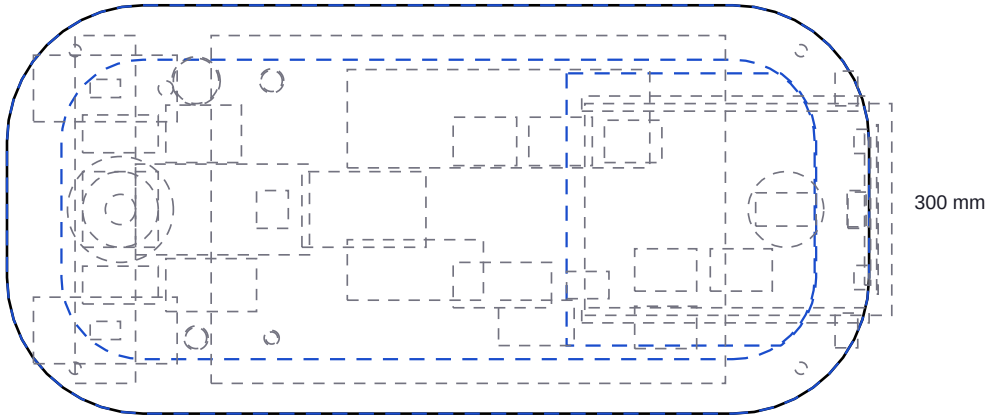
<- must measure 50 mm

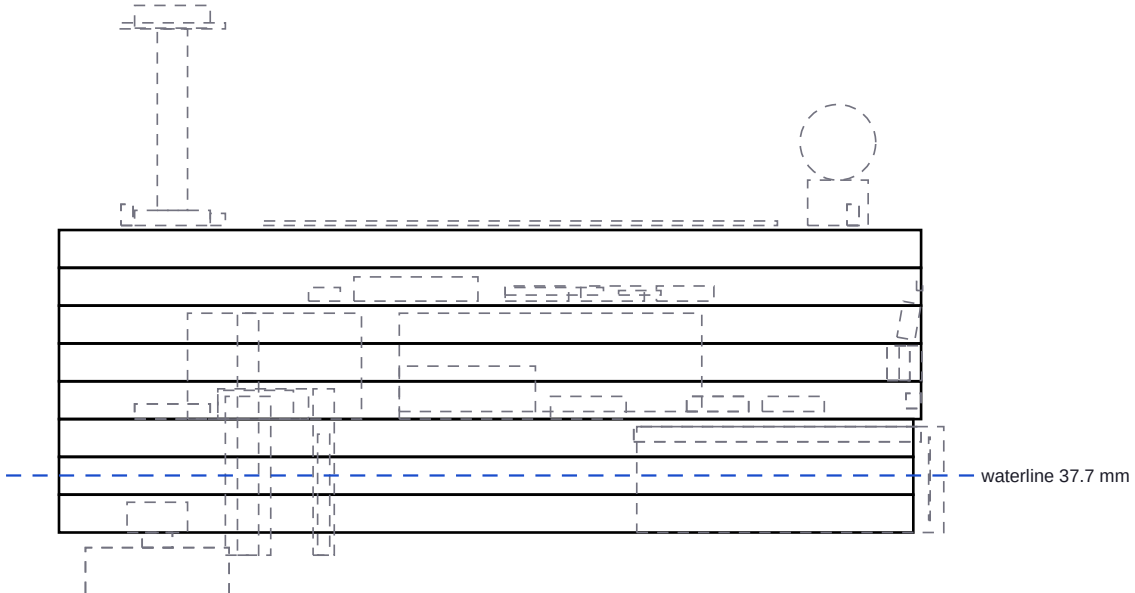
VARUNA AI Pod

600 x 300 mm thermocol boat | Your design, upgraded

PLAN (from above), scale 1:5



SIDE, scale 1:5



Size	600 x 300 mm, hull 200 mm tall, 348 mm to GPS top
Weight	4184 g empty, 4565 g with 381 g of plastic
Draft	37.7 mm empty, 40.9 mm full
Reserve buoyancy	10.5 kg before the deck goes under
Stability (GM)	104 mm: very stable
Trim	bow 2.6 mm higher than stern empty; slide the battery to level
Collection	1.9 L pull-out drawer
Camera	102 mm above water, tilted 10 deg down
Battery	Li-ion 3S6P 20 Ah (222 Wh), about 4.4 h
Thermocol	8 sheets 1000 x 500, or 3 sheets 2 x 3 ft (610 x 915)

How to use these templates

1. Print at 100% / Actual size. Check the 50 mm bar with a ruler.
2. Join tiles by matching the grey 50 mm grid and the tile letters.
3. Stick the paper on thermocol with a glue stick.
4. Cut with a hot-wire cutter or a new snap-off blade, held upright.
5. Stack layers on skewers through the + holes. Glue with Fevicol SH.
6. Never use Fevikwik, spray paint or solvent glue: they melt thermocol.
7. Sand the bow steps smooth, coat with PVA + tissue or packing tape, then water-based acrylic paint.
8. Float-test in a tub before fitting the electronics.

cut line

cut out (window / pocket)

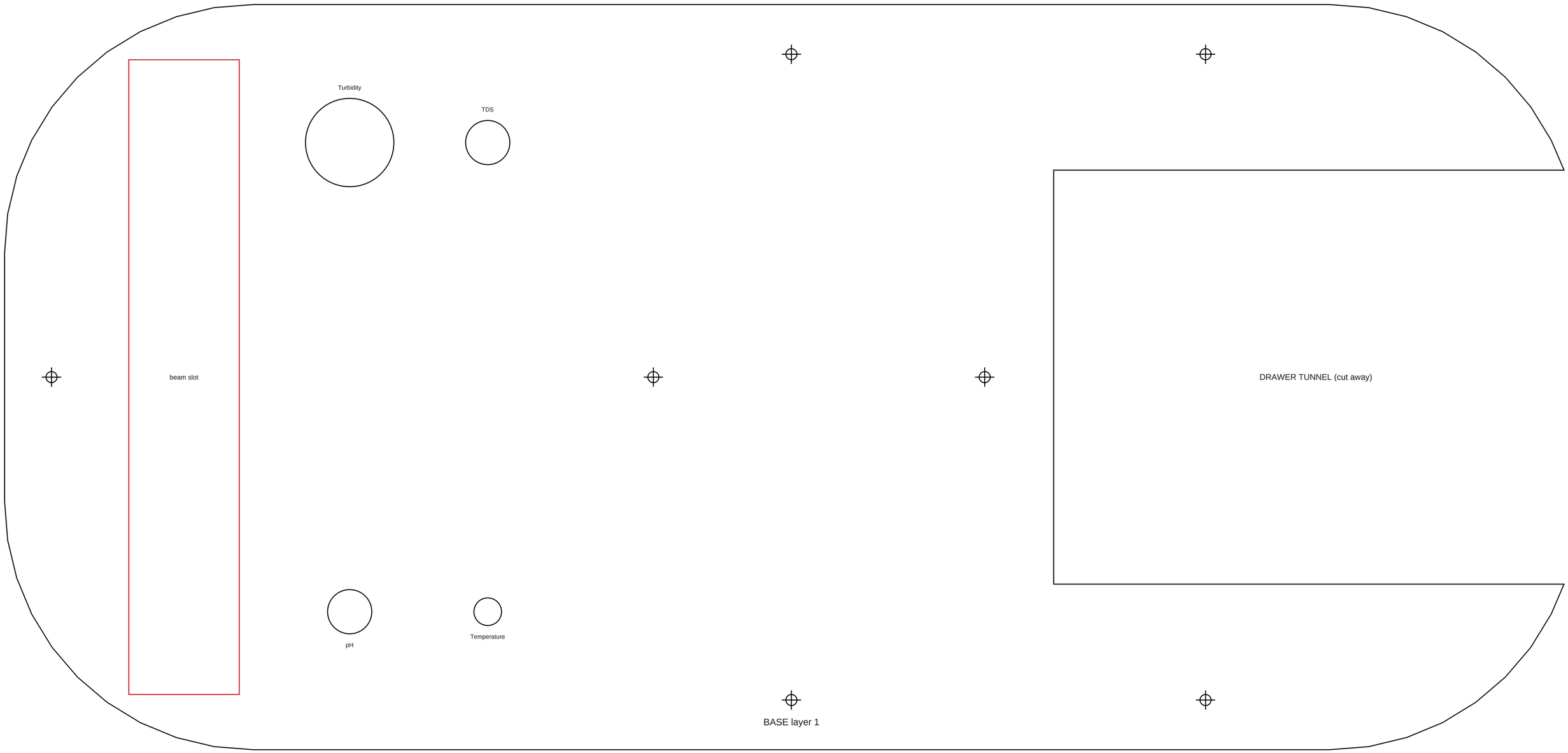
fold / glue line

placement guide, do not cut

skewer hole: lines up the layers

VARUNA AI Pod | Base layer 1 | 25 mm thermocol | A2, print at 100%

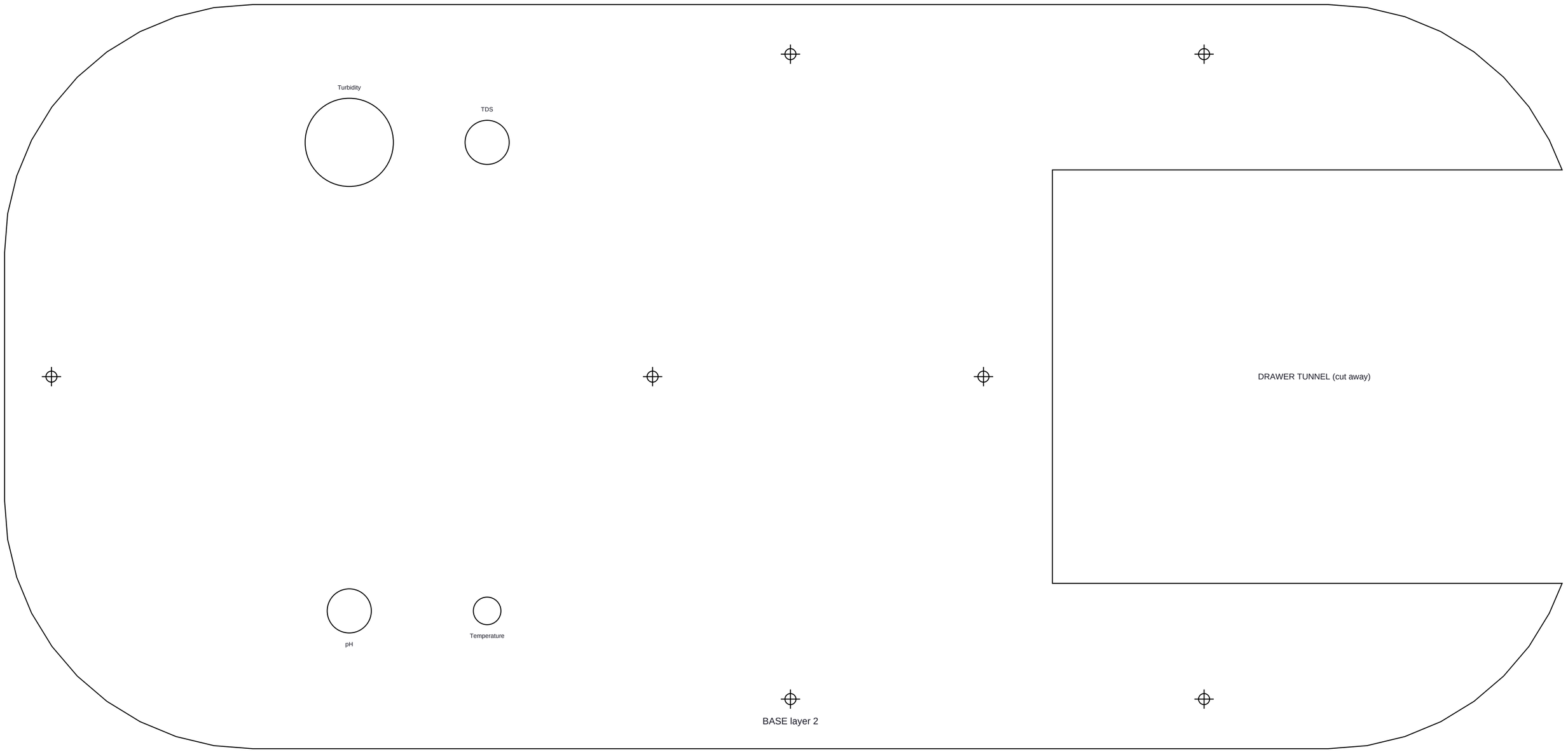
Bow (drawer tunnel) on the right.



<- must measure 100 mm

VARUNA AI Pod | Base layer 2 | 25 mm thermocol | A2, print at 100%

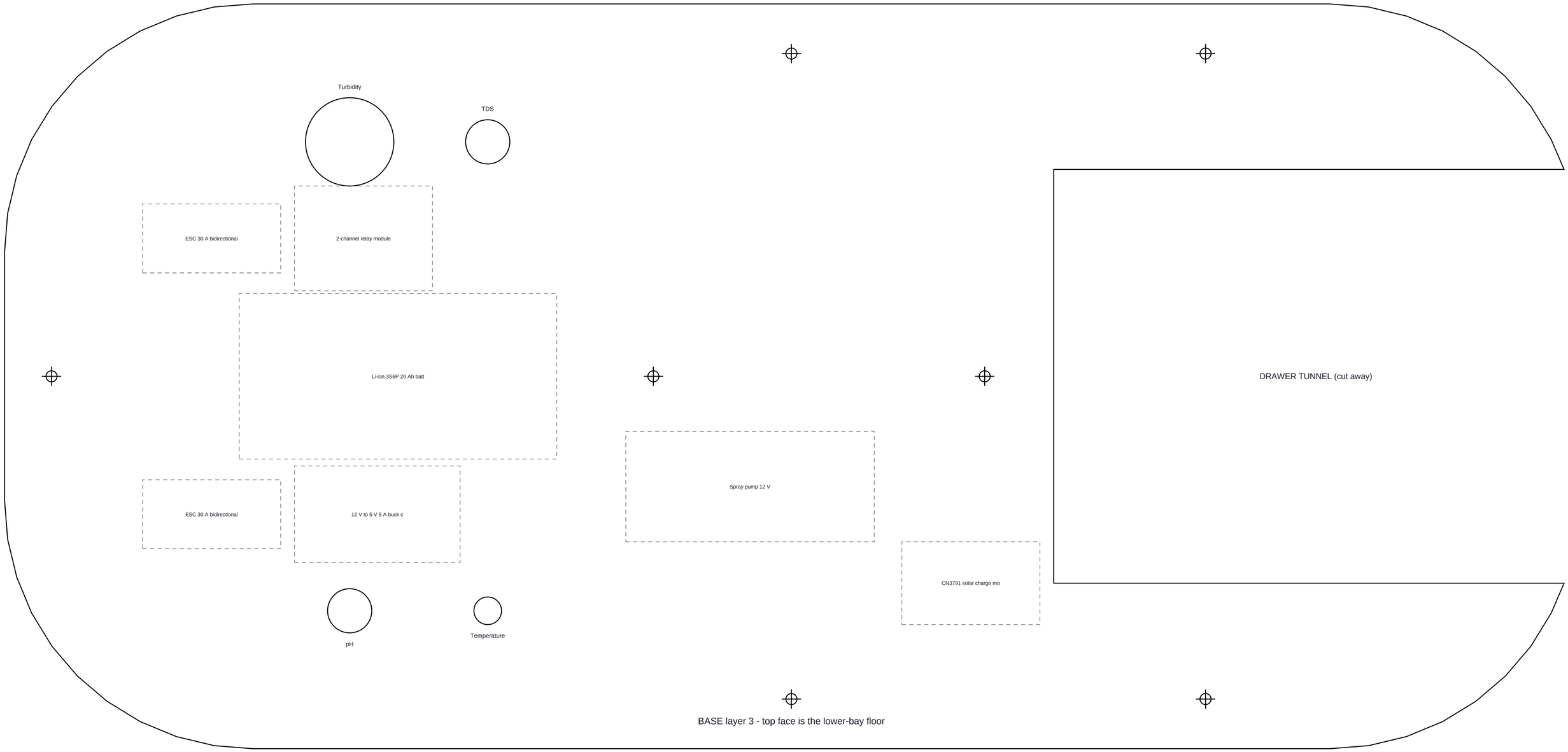
Bow (drawer tunnel) on the right.



<- must measure 100 mm

VARUNA AI Pod | Base layer 3 | 25 mm thermocol | A2, print at 100%

Bow (drawer tunnel) on the right.



<- must measure 100 mm

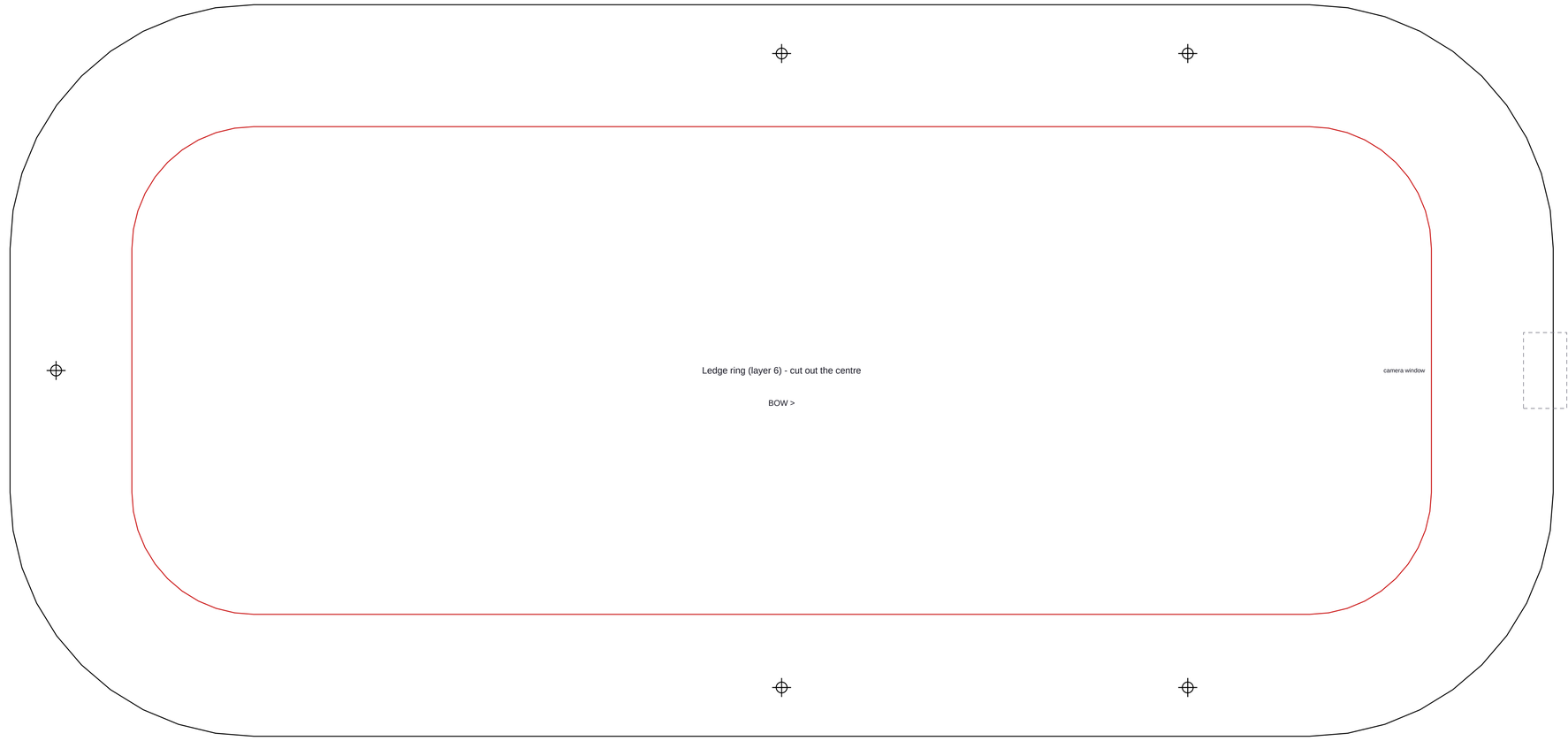
VARUNA AI Pod | Wall ring (layers 4, 5, 7 and lid rim 8) | 25 mm thermocol | A1, print at 100% | PRINT 4 COPIES

Print 4 copies. Layer 5 also gets the 3 ultrasonic holes.

<- must measure 100 mm

VARUNA AI Pod | Ledge ring (layer 6) | 25 mm thermocol | A1, print at 100%

10 mm narrower inside: the shelf rests on it.



<- must measure 100 mm

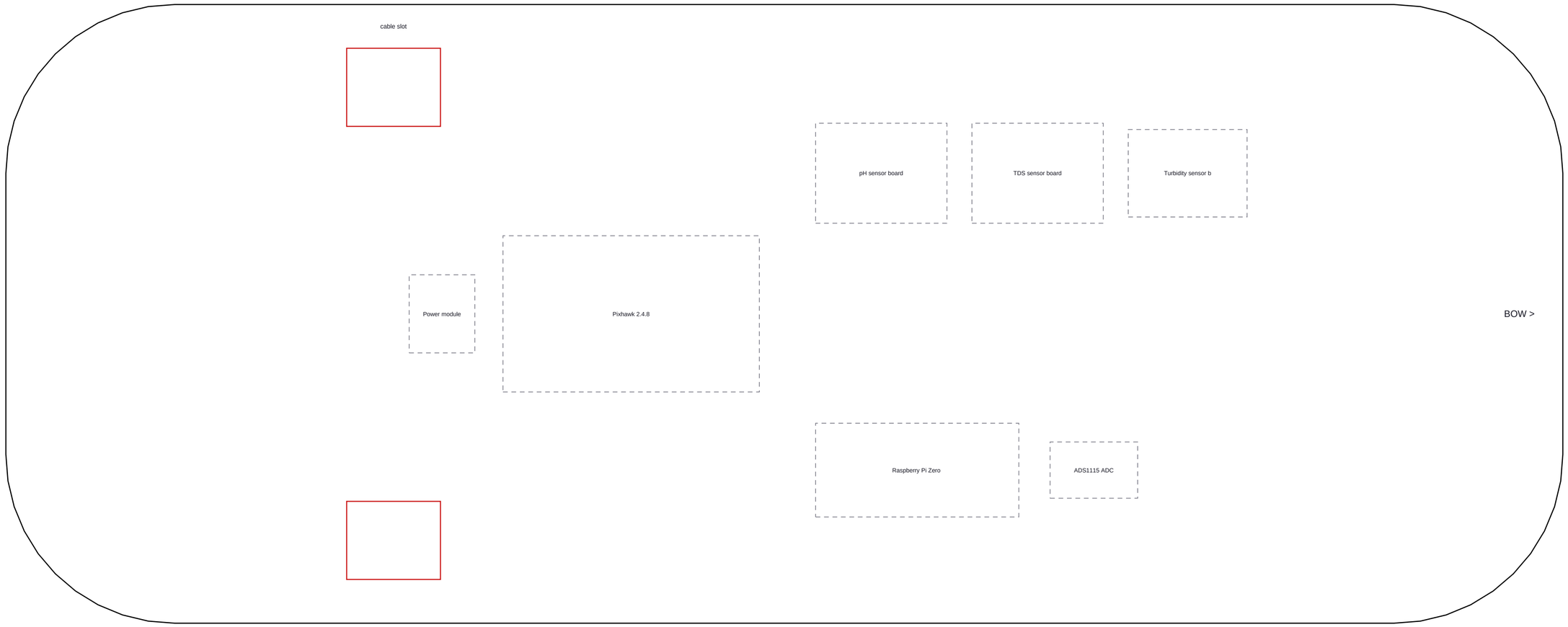
VARUNA AI Pod | Lid top | 3 mm sunboard | A2, print at 100%

Dashed boxes show where the lid parts go.

<- must measure 100 mm

VARUNA AI Pod | Shelf | 3 mm sunboard | A2, print at 100%

Rests on the ledge ring. Dashed boxes: upper-bay parts.

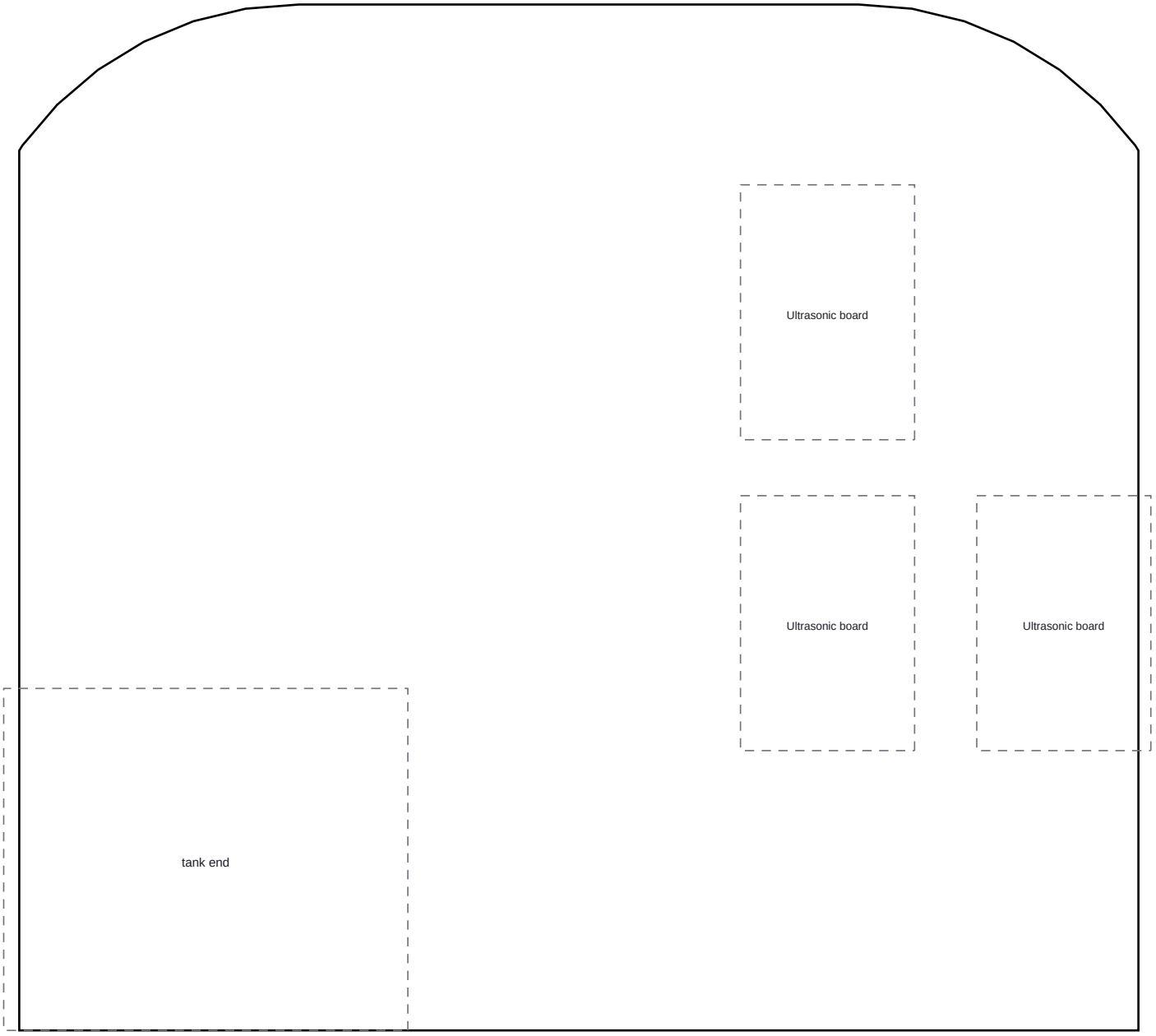


<- must measure 100 mm

VARUNA AI Pod | small parts 1 of 5

<- must measure 50 mm

Drawer tunnel roof
5 mm sunboard

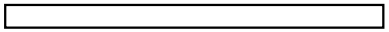


<- must measure 50 mm

Drawer bottom

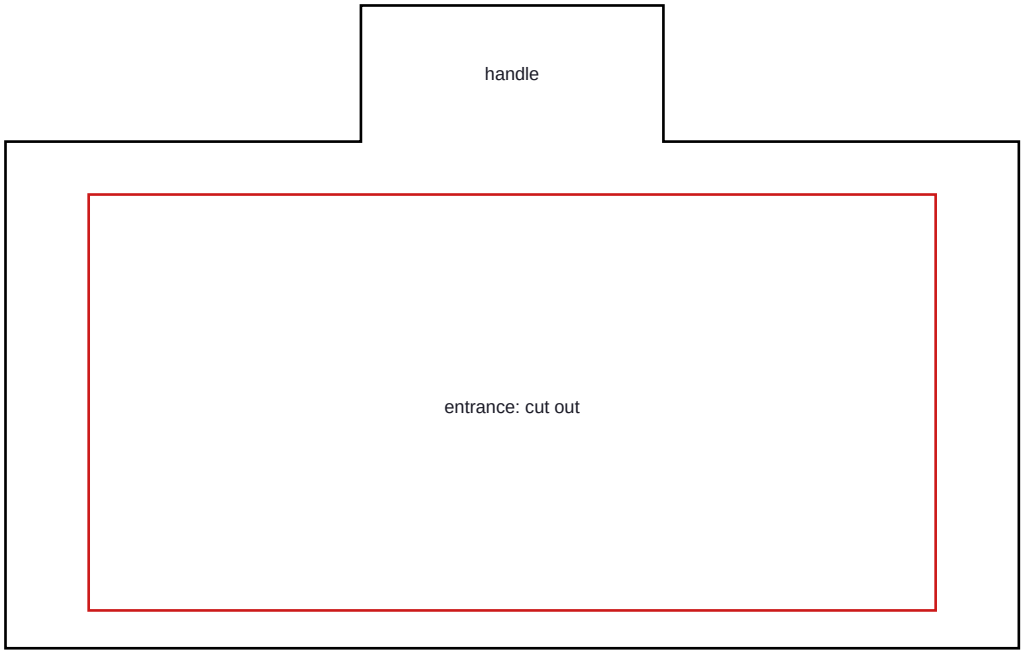
3 mm sunboard

VARUNA AI Pod | small parts 3 of 5

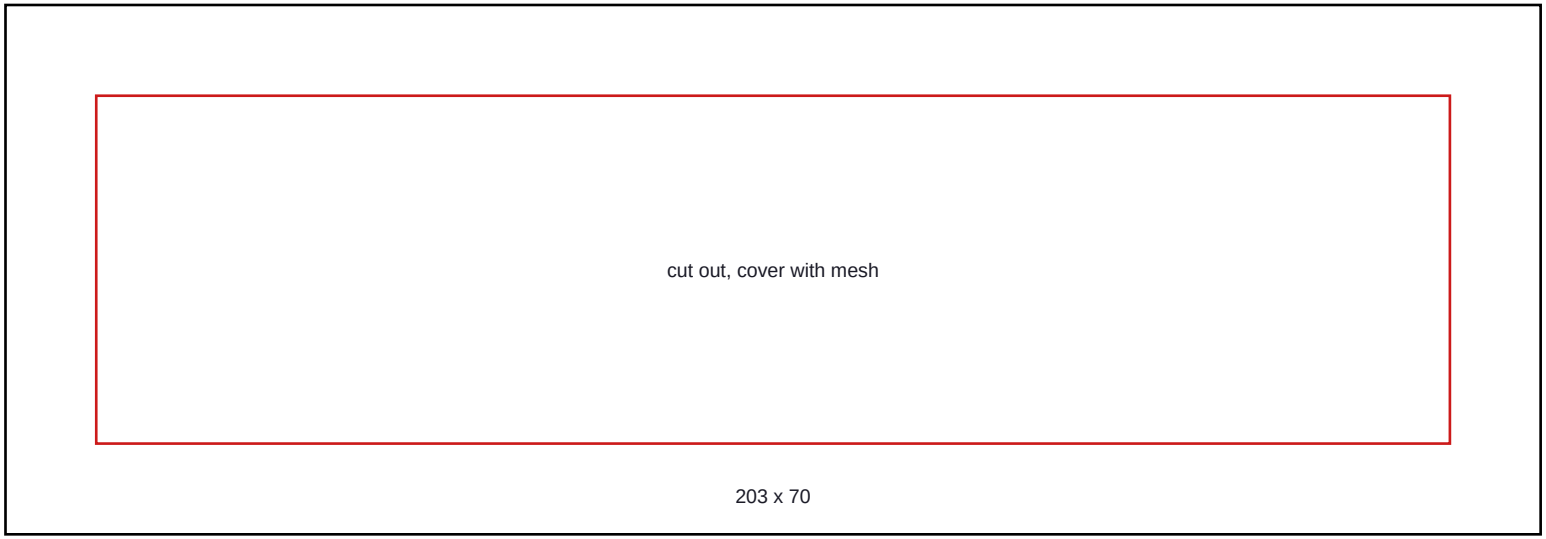


<- must measure 50 mm

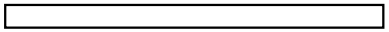
Drawer front (entrance + handle)
3 mm sunboard



Drawer side (left)
3 mm sunboard

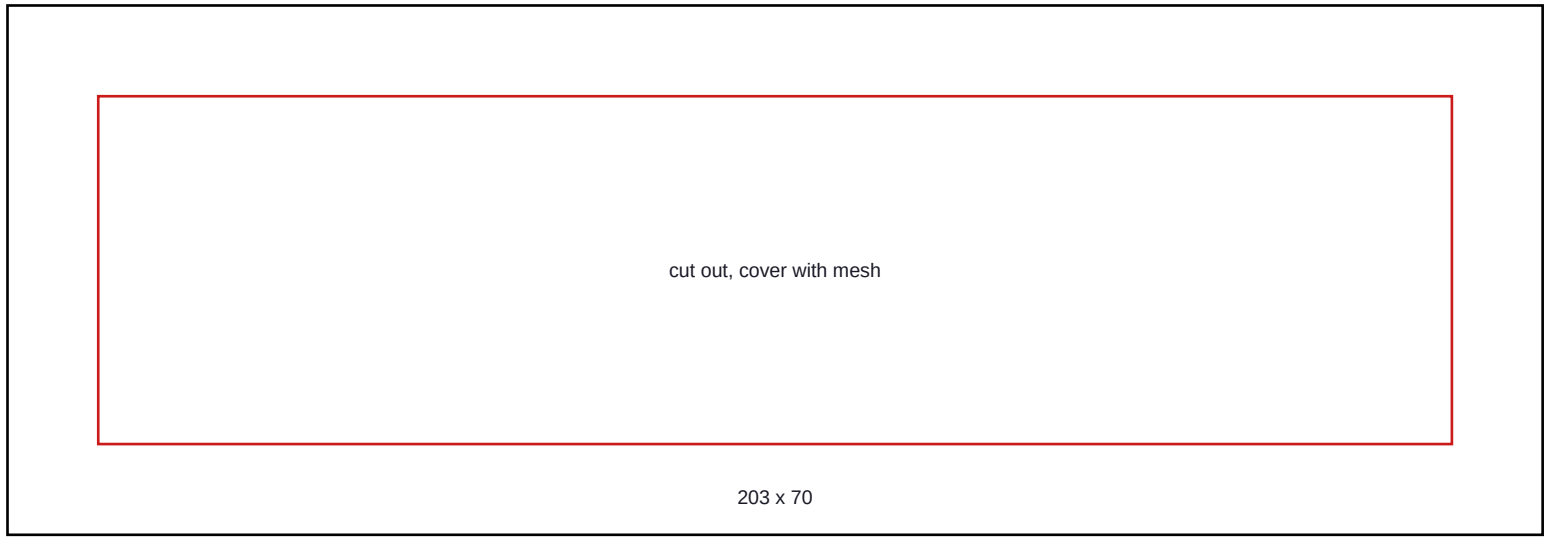


VARUNA AI Pod | small parts 4 of 5

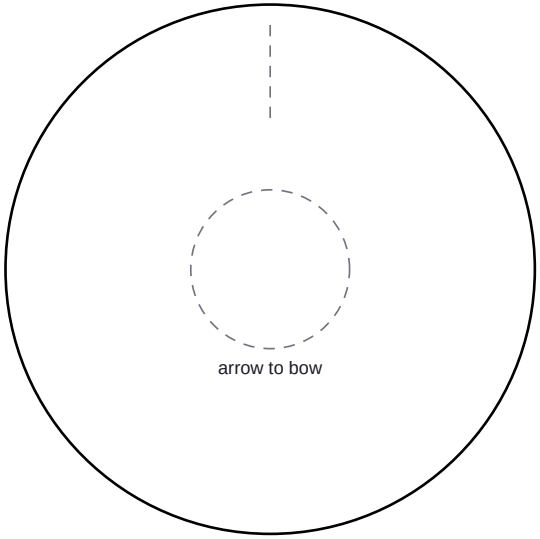


<- must measure 50 mm

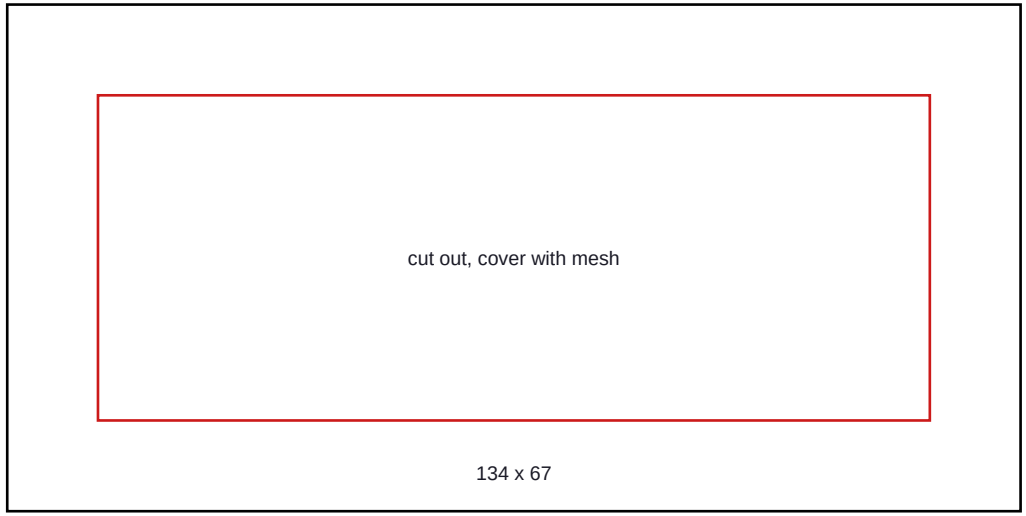
Drawer side (right)
3 mm sunboard



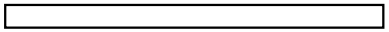
GPS disc
5 mm sunboard



Drawer back
3 mm sunboard

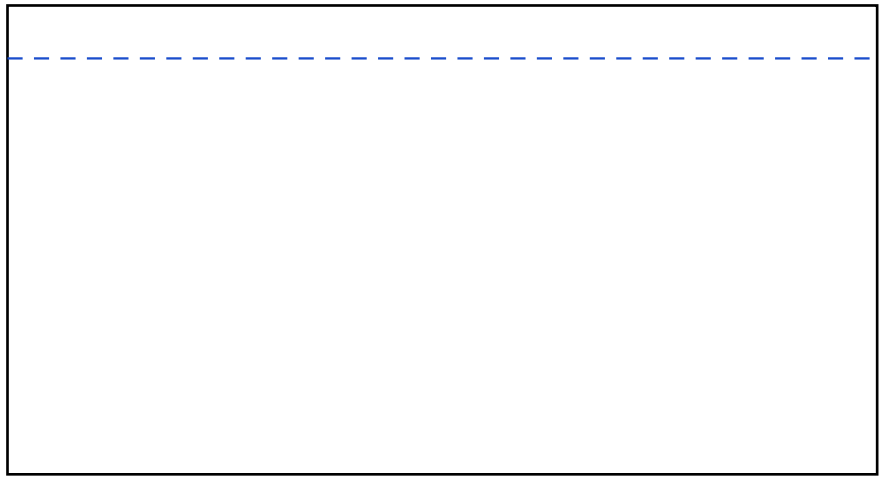


VARUNA AI Pod | small parts 5 of 5

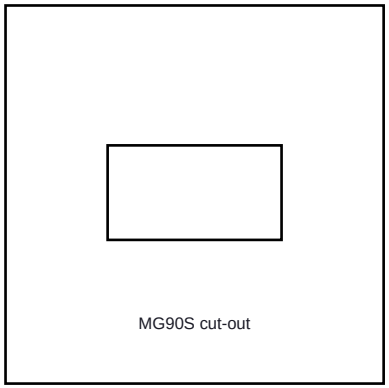


<- must measure 50 mm

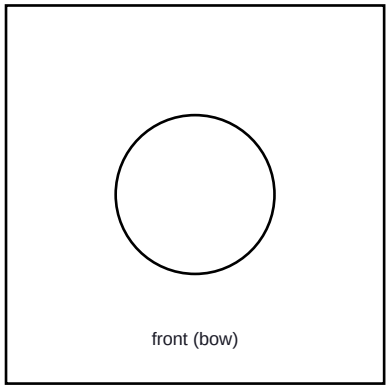
One-way flap
PET sheet



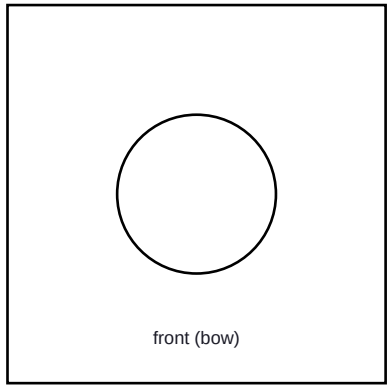
Servo plate
3 mm sunboard



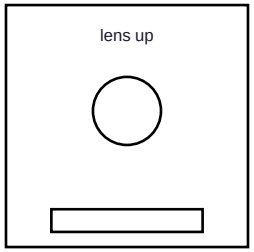
Mast socket layer
5 mm sunboard (cut 2, top one with hole)



Mast socket layer
5 mm sunboard (cut 2, top one with hole)



Camera plate
3 mm sunboard



Camera wedge side (10 deg)
3 mm sunboard

Camera wedge side (10 deg)
3 mm sunboard